

# Week 4 — Probability Review, and the Markowitz Portfolio

AD7031 Optimization Under Uncertainty • Instructor: Prof. H. Park • Sep 19, 2026

## 1. Random variables, events, conditioning

back to week 2's Section 2 table: the OR column had **uncertain data**  $\xi$ . today, the vocabulary — written in this course's notation from the first line —  $x$  is already taken by the **decision**, so the uncertain data is  $\tilde{\xi}$  throughout.

- **random experiment**: its outcome is not known in advance
- **sample space**  $\Omega$  = the set of all possible outcomes

an **event** is a subset of the sample space,  $A \subseteq \Omega$ ;  $\mathbb{P}(A)$  is the probability it occurs.

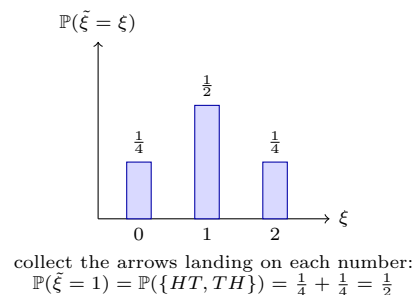
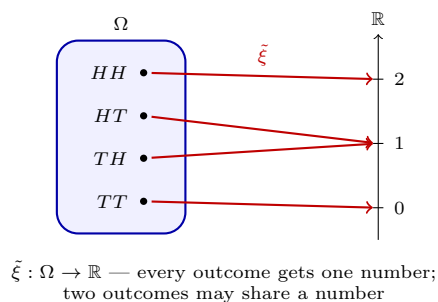
- $0 \leq \mathbb{P}(A) \leq 1$ ,  $\mathbb{P}(\Omega) = 1$
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$  — subtract the overlap, or it is counted twice

**Def** (*random variable*) a function that assigns a real number to each outcome:  $\tilde{\xi} : \Omega \rightarrow \mathbb{R}$

the tilde marks the *random* object;  $\xi$  without it is a **realization** — one value it took

toss a coin twice:  $\Omega = \{HH, HT, TH, TT\}$ , each outcome has probability  $\frac{1}{4}$

$\tilde{\xi}$  = number of heads:  $\tilde{\xi}(HH) = 2$ ,  $\tilde{\xi}(HT) = 1$ ,  $\tilde{\xi}(TH) = 1$ ,  $\tilde{\xi}(TT) = 0$



- **discrete**  $\tilde{\xi}$ : countably many values — number of defective parts, number of arrivals, demand
- **continuous**  $\tilde{\xi}$ : values fill an interval — length, time, temperature, a return
- **support**  $\Xi$  = the set of values  $\tilde{\xi}$  actually takes (the smallest closed set with  $\mathbb{P}(\tilde{\xi} \in \Xi) = 1$ )  
—  $\Omega$  is where the *experiment* lives,  $\Xi \subseteq \mathbb{R}^k$  is where the *numbers* live

**Def** (*conditional probability*)  $\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$ , defined when  $\mathbb{P}(B) > 0$

—  $B$  becomes the new sample space; divide by  $\mathbb{P}(B)$  so the weights add to 1 again

two coins:  $A = \{\text{both flips are heads}\} = \{HH\}$ ,  $B = \{\text{first flip is head}\} = \{HH, HT\}$

$$\mathbb{P}(A) = \frac{1}{4}, \quad \mathbb{P}(B) = \frac{1}{2}, \quad A \cap B = \{HH\}: \quad \mathbb{P}(A | B) = \frac{1/4}{1/2} = \frac{1}{2}$$

— the first head doubles it,  $\frac{1}{4} \rightarrow \frac{1}{2}$

**Def** (*independent events*)  $A$  and  $B$  are independent if  $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$

then  $\mathbb{P}(A | B) = \frac{\mathbb{P}(A)\mathbb{P}(B)}{\mathbb{P}(B)} = \mathbb{P}(A)$  — learning  $B$  tells you nothing about  $A$

same coins, same  $B$ . now  $A = \{\text{exactly one head}\} = \{HT, TH\}$ :

$$\mathbb{P}(A \cap B) = \mathbb{P}(\{HT\}) = \frac{1}{4}, \quad \mathbb{P}(A) \mathbb{P}(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \quad \text{equal — independent}$$

so  $\mathbb{P}(A | B) = \frac{1}{2} = \mathbb{P}(A)$ : the first flip says nothing about whether there is exactly one head

now run the same test on the *first* pair,  $A = \{\text{both heads}\} = \{HH\}$ :

$$\mathbb{P}(A \cap B) = \mathbb{P}(\{HH\}) = \frac{1}{4}, \quad \mathbb{P}(A) \mathbb{P}(B) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8} \quad \frac{1}{4} \neq \frac{1}{8} \text{ — dependent}$$

which is why conditioning moved it there:  $\mathbb{P}(A | B) = \frac{1}{2} \neq \frac{1}{4} = \mathbb{P}(A)$

same  $B$ , two different  $A$ , opposite answers — independence is a property of the *pair*, not of either event on its own.

**Thm (law of total probability)** let  $A_1, \dots, A_I$  **partition**  $\Omega$ :  $A_i \cap A_j = \emptyset$  for  $i \neq j$ , and  $\bigcup_i A_i = \Omega$ . then for any event  $B$ ,

$$\mathbb{P}(B) = \mathbb{P}(B \cap A_1) + \dots + \mathbb{P}(B \cap A_I) = \mathbb{P}(B | A_1) \mathbb{P}(A_1) + \dots + \mathbb{P}(B | A_I) \mathbb{P}(A_I)$$

— split the world into cases, answer each case, weight by how likely the case is

two coins again:  $A_1 = B$  (first flip head),  $A_2 = B^c$  (first flip tail) — a partition, each  $\frac{1}{2}$

the event of interest: *at least one head*  $\mathbb{P}(\text{at least one head} | A_1) = 1$  (a head already landed),  $\mathbb{P}(\text{at least one head} | A_2) = \frac{1}{2}$  (needs the second)

$$\mathbb{P}(\text{at least one head}) = 1 \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} \quad \text{— the same } \frac{3}{4} \text{ Section 2 gives from the scenario list}$$

**takeaway** conditioning is how a *two-stage* decision gets written down: act, watch which  $A_i$  occurred, act again. weeks 10–12 run on this.

## 2. Discrete random variables

a discrete  $\tilde{\xi}$  is a **list**: finitely many values, each carrying a weight.

**Def (scenarios)**  $\tilde{\xi}$  takes the values  $\xi^1, \dots, \xi^S$  — the **scenarios** — with

$$p_s = \mathbb{P}(\tilde{\xi} = \xi^s), \quad (1) \ p_s \geq 0 \quad (2) \ \sum_{s \in [S]} p_s = 1, \quad [S] = \{1, \dots, S\}$$

$\mathbb{P}(\tilde{\xi} \in A) = \sum_{s: \xi^s \in A} p_s$  — add the weights of the scenarios landing in  $A$

**the scenario index goes up**:  $\xi^s$  is scenario  $s$ ; a *subscript*  $\xi_j$  is component  $j$  of a vector (Section 7). two different jobs, two different slots.

older books call this list the **probability mass function** and write it as a function  $f$ . same content — but a table, and a table is the only thing a solver can be handed.

$$\text{two coins: } S = 3, \quad \xi^1 = 0, \ \xi^2 = 1, \ \xi^3 = 2, \quad p = \left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right)$$

$$\mathbb{P}(\tilde{\xi} \geq 1) = p_2 + p_3 = \frac{3}{4} \quad \text{— the } \frac{3}{4} \text{ Section 1 reached by conditioning}$$

**Def (mean)**  $\mu = \mathbb{E}[\tilde{\xi}] = \sum_{s \in [S]} p_s \xi^s$

function of  $\tilde{\xi}$ :  $\mathbb{E}[h(\tilde{\xi})] = \sum_{s \in [S]} p_s h(\xi^s)$   $\leftarrow$  law of the unconscious statistician

— keep the weights  $p_s$ , change only what they weight. no new distribution needed

**Linearity of expectation**  $\mathbb{E}$  is a weighted sum, so constants pull out and sums split:

$$\mathbb{E}[c] = c, \quad \mathbb{E}[a\tilde{\xi} + b] = a \mathbb{E}[\tilde{\xi}] + b, \quad \mathbb{E}[g(\tilde{\xi}) + h(\tilde{\xi})] = \mathbb{E}[g(\tilde{\xi})] + \mathbb{E}[h(\tilde{\xi})]$$

— each one is  $\sum_s p_s(\dots)$  pulled apart; there is nothing deeper in it

**Def (variance)**  $\sigma^2 = \text{Var}(\tilde{\xi}) = \mathbb{E}[(\tilde{\xi} - \mu)^2] = \sum_{s \in [S]} p_s (\xi^s - \mu)^2, \quad \sigma = \sqrt{\text{Var}(\tilde{\xi})}$

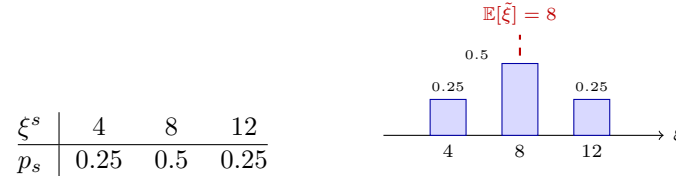
the **shortcut** — expand the square, then apply linearity term by term:

$$\begin{aligned}\text{Var}(\tilde{\xi}) &= \mathbb{E}[(\tilde{\xi} - \mu)^2] = \mathbb{E}[\tilde{\xi}^2 - 2\mu\tilde{\xi} + \mu^2] \\ &= \mathbb{E}[\tilde{\xi}^2] - 2\mu\mathbb{E}[\tilde{\xi}] + \mu^2 = \mathbb{E}[\tilde{\xi}^2] - 2\mu^2 + \mu^2 = \mathbb{E}[\tilde{\xi}^2] - \mu^2\end{aligned}$$

$\mu$  is a number, not random — that is what lets it come out of the  $\mathbb{E}$

$$\text{Var}(a\tilde{\xi} + b) = a^2 \text{Var}(\tilde{\xi}) \quad \text{— shifting by } b \text{ moves the centre, not the spread}$$

**Example 1** demand  $\tilde{\xi}$  for a spare part next month:



$$\mathbb{E}[\tilde{\xi}] = 0.25 \cdot 4 + 0.5 \cdot 8 + 0.25 \cdot 12 = 8$$

$$\text{Var}(\tilde{\xi}) = 0.25 \cdot (4 - 8)^2 + 0.5 \cdot (8 - 8)^2 + 0.25 \cdot (12 - 8)^2 = 8, \quad \sigma = 2\sqrt{2} \approx 2.83$$

### 3. Continuous random variables

**Def** (*probability density function, PDF*)  $f : \mathbb{R} \rightarrow \mathbb{R}_+$  with

$$(1) \ p(\xi) \geq 0 \quad (2) \ \int_{-\infty}^{\infty} p(\xi) d\xi = 1 \quad (3) \ \mathbb{P}(a \leq \tilde{\xi} \leq b) = \int_a^b p(\xi) d\xi = \text{area under } p \text{ from } a \text{ to } b$$

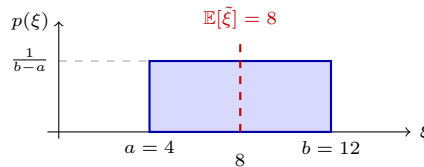
$p(\xi)$  is *not* a probability:  $\mathbb{P}(\tilde{\xi} = \xi) = 0$  for every single  $\xi$ , and  $p(\xi) > 1$  is allowed

**Def** (*mean, variance*) same as discrete — the sums become integrals:

$$\mu = \mathbb{E}[\tilde{\xi}] = \int \xi p(\xi) d\xi, \quad \sigma^2 = \text{Var}(\tilde{\xi}) = \int (\xi - \mu)^2 p(\xi) d\xi = \mathbb{E}[\tilde{\xi}^2] - \mu^2, \quad \mathbb{E}[h(\tilde{\xi})] = \int h(\xi) p(\xi) d\xi$$

the shortcut  $\text{Var}(\tilde{\xi}) = \mathbb{E}[\tilde{\xi}^2] - \mu^2$  carries over unchanged: Section 2 derived it from linearity alone, and linearity holds here for the same reason — the integral splits and constants come out

**Example 2**  $\tilde{\xi} \sim \text{Uniform}[a, b]$  —  $p(\xi) = \frac{1}{b-a}$  on  $[a, b]$ , 0 outside



mean:

$$\mathbb{E}[\tilde{\xi}] = \int_a^b \xi \frac{1}{b-a} d\xi = \frac{1}{b-a} \left[ \frac{x^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}$$

second moment —  $h(x) = x^2$ :

$$\mathbb{E}[\tilde{\xi}^2] = \int_a^b \xi^2 \frac{1}{b-a} d\xi = \frac{1}{b-a} \left[ \frac{x^3}{3} \right]_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{a^2 + ab + b^2}{3}$$

variance — subtract, no new integral:

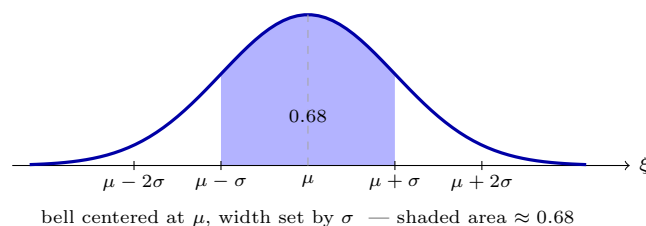
$$\text{Var}(\tilde{\xi}) = \mathbb{E}[\tilde{\xi}^2] - \mathbb{E}[\tilde{\xi}]^2 = \frac{a^2 + ab + b^2}{3} - \frac{(a+b)^2}{4} = \frac{(b-a)^2}{12}$$

$$\text{with } a = 4, b = 12: \quad \mathbb{E}[\tilde{\xi}] = 8, \quad \text{Var}(\tilde{\xi}) = \frac{64}{12} = \frac{16}{3}, \quad \sigma = \frac{4}{\sqrt{3}} \approx 2.31$$

**Normal distribution**  $\tilde{\xi} \sim \mathcal{N}(\mu, \sigma^2)$ :  $p(\xi) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\xi - \mu)^2}{2\sigma^2}\right)$ ,  $\xi \in \mathbb{R}$

$$\mathbb{E}[\tilde{\xi}] = \mu, \quad \text{Var}(\tilde{\xi}) = \sigma^2$$

$$\mathbb{P}(\mu - \sigma \leq \tilde{\xi} \leq \mu + \sigma) \approx 0.68, \quad \mathbb{P}(\mu - 2\sigma \leq \tilde{\xi} \leq \mu + 2\sigma) \approx 0.95$$



## 4. Expectation, in one formula

Sections 2 and 3 defined the mean twice — once a sum, once an integral. one definition covers both.

**Def (expectation)** for  $\tilde{\xi} \sim \mathbb{P}$ ,

$$\mathbb{E}_{\mathbb{P}}[\tilde{\xi}] = \int_{\Omega} \xi \mathbb{P}(d\xi)$$

read  $\mathbb{P}(d\xi)$  as *the probability sitting on a small piece  $d\xi$*  — the weight, whichever form it takes:

|            | the weight $\mathbb{P}(d\xi)$ is | and $\mathbb{E}_{\mathbb{P}}[\tilde{\xi}]$ reads |
|------------|----------------------------------|--------------------------------------------------|
| discrete   | $p_s$ , one per scenario $\xi^s$ | $\sum_{s \in [S]} p_s \xi^s$                     |
| continuous | $p(\xi) d\xi$                    | $\int_{\Xi} \xi p(\xi) d\xi$                     |

**Def (expectation of a function)** for  $f: \mathbb{R}^k \rightarrow \mathbb{R}$  and a random vector  $\tilde{\xi}$ ,

$$\mathbb{E}_{\mathbb{P}}[f(\tilde{\xi})] = \int_{\Omega} f(\xi) \mathbb{P}(d\xi) \quad \text{—} \quad \sum_{s \in [S]} p_s f(\xi^s) \quad \text{or} \quad \int_{\Xi} f(\xi) p(\xi) d\xi$$

this is the one the course actually needs:  $f$  will be a **cost**, and  $\mathbb{E}[f(x, \tilde{\xi})]$  the objective.

variance is an expectation too, so it inherits the same two readings:

$$\text{Var}(\tilde{\xi}) = \mathbb{E}[(\tilde{\xi} - \mathbb{E}[\tilde{\xi}])^2], \quad \sigma(\tilde{\xi}) = \sqrt{\text{Var}(\tilde{\xi})}, \quad \text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2) = \mathbb{E}[(\tilde{\xi}_1 - \mathbb{E}[\tilde{\xi}_1])(\tilde{\xi}_2 - \mathbb{E}[\tilde{\xi}_2])]$$

**takeaway** one definition, two computations. write  $\mathbb{E}$  once and stop asking whether the data is discrete or continuous.

## 5. Two random variables

**Def (joint PMF)**  $p(\xi_1, \xi_2) = \mathbb{P}(\tilde{\xi}_1 = \xi_1, \tilde{\xi}_2 = \xi_2)$ ,  $p \geq 0$ ,  $\sum_x \sum_y p(\xi_1, \xi_2) = 1$

**Def (marginal PMF)**  $p_1(\xi_1) = \sum_y p(\xi_1, \xi_2)$ ,  $p_2(\xi_2) = \sum_x p(\xi_1, \xi_2)$

— row sums and column sums, written in the *margins* of the table

**Example 3** two assets, returns tomorrow:  $\tilde{\xi}_1, \tilde{\xi}_2$  each either down ( $-0.10$ ) or up ( $+0.20$ )

| $p(\xi_1, \xi_2)$ | $\xi_2 = -0.10$ | $\xi_2 = 0.20$ | $p_1(\xi_1)$ |
|-------------------|-----------------|----------------|--------------|
| $\xi_1 = -0.10$   | 0.3             | 0.2            | 0.5          |
| $\xi_1 = 0.20$    | 0.2             | 0.3            | 0.5          |
| $p_2(\xi_2)$      | 0.5             | 0.5            | 1            |

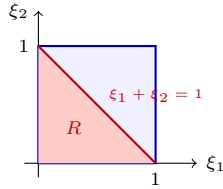
each asset alone: down or up w.p.  $\frac{1}{2}$ .    together: same direction w.p.  $0.3 + 0.3 = 0.6$

**Def (joint PDF)**  $p : \mathbb{R}^2 \rightarrow \mathbb{R}_+$ ,  $\iint p(\xi_1, \xi_2) d\xi_1 d\xi_2 = 1$ ,  $\mathbb{P}((\tilde{\xi}_1, \tilde{\xi}_2) \in R) = \iint_R p(\xi_1, \xi_2) d\xi_1 d\xi_2$

— the **volume** under  $p$  above the region  $R$

marginal PDF:  $p_1(\xi_1) = \int p(\xi_1, \xi_2) d\xi_2$  — the sum becomes an integral

**Example 4**  $(\tilde{\xi}_1, \tilde{\xi}_2)$  uniform on the unit square:  $p(\xi_1, \xi_2) = 1$  on  $[0, 1]^2$ , 0 outside



$$\mathbb{P}(\tilde{\xi}_1 + \tilde{\xi}_2 \leq 1) = \iint_R 1 d\xi_1 d\xi_2 = \text{area of the triangle } R = \frac{1}{2}$$

$$\text{marginal: } p_1(\xi_1) = \int_0^1 1 d\xi_2 = 1 \text{ on } [0, 1] \text{ — } \tilde{\xi}_1 \text{ alone is Uniform}[0, 1]$$

**Def (independence)**  $\tilde{\xi}_1, \tilde{\xi}_2$  are independent if  $p(\xi_1, \xi_2) = p_1(\xi_1)p_2(\xi_2) \quad \forall \xi_1, \xi_2$  (PMF or PDF)

Example 3:  $p(-0.10, -0.10) = 0.3 \neq 0.5 \cdot 0.5$  — **not** independent

Example 4:  $p(\xi_1, \xi_2) = 1 = 1 \cdot 1$  — independent

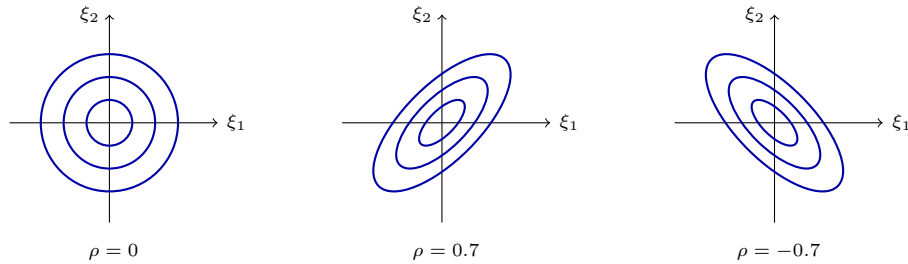
## 6. Covariance and correlation

**Def (covariance, correlation)**

$$\text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2) = \mathbb{E}[(\tilde{\xi}_1 - \mu_1)(\tilde{\xi}_2 - \mu_2)] = \mathbb{E}[\tilde{\xi}_1 \tilde{\xi}_2] - \mu_1 \mu_2, \quad \text{Cov}(\tilde{\xi}_1, \tilde{\xi}_1) = \text{Var}(\tilde{\xi}_1)$$

$$\rho_{12} = \frac{\text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2)}{\sigma_1 \sigma_2}, \quad -1 \leq \rho_{12} \leq 1$$

$\rho > 0$ : move together,  $\rho < 0$ : move opposite, independent  $\Rightarrow \rho = 0$



contours of the joint PDF of two normal random variables,  $\mu_1 = \mu_2 = 0$ ,  $\sigma_1 = \sigma_2 = 1$

**Example 3, continued** (use the margins for  $\mu$  and  $\sigma$ , all four cells for the covariance)

$$\mu_1 = 0.5 \cdot (-0.10) + 0.5 \cdot 0.20 = 0.05, \quad \mu_2 = 0.05$$

$$\text{Var}(\tilde{\xi}_1) = 0.5 \cdot (-0.15)^2 + 0.5 \cdot (0.15)^2 = 0.0225, \quad \sigma_1 = \sigma_2 = 0.15$$

$$\begin{aligned} \text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2) &= 0.3(-0.15)(-0.15) + 0.2(-0.15)(0.15) + 0.2(0.15)(-0.15) + 0.3(0.15)(0.15) \\ &= (0.3 - 0.2 - 0.2 + 0.3) \cdot 0.0225 = 0.0045 \end{aligned}$$

$$\rho_{12} = \frac{0.0045}{0.15 \cdot 0.15} = 0.2 \quad \text{— the two assets tend to move together}$$

## 7. Linear combinations

$$\mathbb{E}[a\tilde{\xi}_1 + b\tilde{\xi}_2] = a\mu_1 + b\mu_2 \quad \text{— always, no independence needed}$$

$$\text{Var}(a\tilde{\xi}_1 + b\tilde{\xi}_2) = a^2 \text{Var}(\tilde{\xi}_1) + b^2 \text{Var}(\tilde{\xi}_2) + 2ab \text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2)$$

$n$  random variables  $\tilde{\xi} = (\tilde{\xi}_1, \dots, \tilde{\xi}_n)$ :

mean vector  $\mu = (\mathbb{E}[\tilde{\xi}_1], \dots, \mathbb{E}[\tilde{\xi}_n])^\top \in \mathbb{R}^n$

covariance matrix  $\Sigma \in \mathbb{R}^{n \times n}$ ,  $\Sigma_{ij} = \text{Cov}(\tilde{\xi}_i, \tilde{\xi}_j)$ ; diagonal  $\Sigma_{ii} = \text{Var}(\tilde{\xi}_i)$ ; symmetric; always  $\Sigma \succeq 0$

for fixed  $c \in \mathbb{R}^n$ :  $\mathbb{E}[c^\top \tilde{\xi}] = c^\top \mu$ ,  $\text{Var}(c^\top \tilde{\xi}) = c^\top \Sigma c$

$n = 2$ ,  $c = (a, b)$ :  $c^\top \Sigma c = \begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} \text{Var}(\tilde{\xi}_1) & \text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2) \\ \text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2) & \text{Var}(\tilde{\xi}_2) \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \text{Var}(a\tilde{\xi}_1 + b\tilde{\xi}_2)$  again

Example 3:  $\mu = \begin{pmatrix} 0.05 \\ 0.05 \end{pmatrix}$ ,  $\Sigma = \begin{pmatrix} 0.0225 & 0.0045 \\ 0.0045 & 0.0225 \end{pmatrix}$

## 8. The bridge to optimization

everything so far was about  $\tilde{\xi}$  alone. now put the decision back in.

a cost depends on both:  $f(x, \tilde{\xi})$  —  $x$  the **decision** we choose,  $\tilde{\xi}$  the **data** we do not

fix  $x$ . then  $f(x, \tilde{\xi})$  is a **random variable** — it inherits  $\tilde{\xi}$ 's randomness

so  $\mathbb{E}[f(x, \tilde{\xi})]$  is a **number**, one per  $x$  — and numbers can be minimized:

$$\min_{x \in X} \mathbb{E}[f(x, \tilde{\xi})]$$

that  $\mathbb{E}$  is the whole design space of this course. expectation is one way to turn a random cost into a number; CVaR, VaR and a chance constraint are others. the model above changes in that one slot and nowhere else.

**takeaway** every paradigm in this course differs in one place only: *how it collapses the randomness to a number.*

## 9. Running example: the Markowitz portfolio

a budget of 1,  $n$  assets. decide the split today; the returns are revealed tomorrow.

- **decision:**  $x \in \mathbb{R}^n$ ,  $x_j$  = share of the budget in asset  $j$
- **uncertain data:**  $\tilde{\xi} = (\tilde{\xi}_1, \dots, \tilde{\xi}_n) \in \mathbb{R}^n$ ,  $\tilde{\xi}_j$  = return of asset  $j$  — a random vector with mean  $\mu$ , covariance matrix  $\Sigma$
- **feasible set:**  $X = \{x \in \mathbb{R}^n : \mathbf{1}^\top x = 1, x \geq 0\}$  ( $\mathbf{1}$  = all-ones vector,  $\mathbf{1}^\top x = x_1 + \dots + x_n$ )

**Portfolio return**  $\tilde{\xi}^\top x = x_1 \tilde{\xi}_1 + \dots + x_n \tilde{\xi}_n$  — a random variable

as a function: fix  $x$ ,  $f(\xi) = \xi^\top x$ ,  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  — the week 2 note's inner product, fed a random input

loss:  $f(x, \tilde{\xi}) = -\tilde{\xi}^\top x$  (a gain is a negative loss)

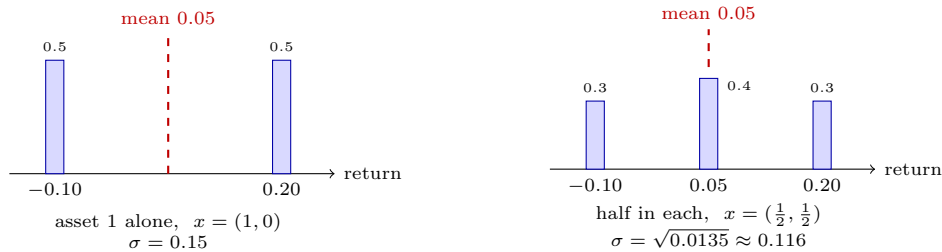
Section 7's linear combinations with  $c = x$ :  $\mathbb{E}[\tilde{\xi}^\top x] = \mu^\top x$ ,  $\text{Var}(\tilde{\xi}^\top x) = x^\top \Sigma x$

**Check on Example 3**  $(\tilde{\xi}_1, \tilde{\xi}_2) = \text{Example 3's } (\tilde{\xi}_1, \tilde{\xi}_2)$ , half in each:  $x = (\frac{1}{2}, \frac{1}{2})$

| $\xi_1$ | $\xi_2$ | $\frac{1}{2}\xi_1 + \frac{1}{2}\xi_2$ | probability | $\Rightarrow$ PMF of $\tilde{\xi}^\top x$ : | value | probability | -0.10 | 0.05 | 0.20 |
|---------|---------|---------------------------------------|-------------|---------------------------------------------|-------|-------------|-------|------|------|
| -0.10   | -0.10   | -0.10                                 | 0.3         |                                             |       |             |       |      |      |
| -0.10   | 0.20    | 0.05                                  | 0.2         |                                             |       |             |       |      |      |
| 0.20    | -0.10   | 0.05                                  | 0.2         |                                             |       |             |       |      |      |
| 0.20    | 0.20    | 0.20                                  | 0.3         |                                             |       |             |       |      |      |

from the PMF:  $\mathbb{E}[\tilde{\xi}^\top x] = 0.3(-0.10) + 0.4(0.05) + 0.3(0.20) = 0.05$   
 $\text{Var}(\tilde{\xi}^\top x) = 0.3(-0.15)^2 + 0.4(0)^2 + 0.3(0.15)^2 = 0.0135$

from  $\mu$  and  $\Sigma$ :  $\mu^\top x = \frac{1}{2}(0.05) + \frac{1}{2}(0.05) = 0.05 \checkmark$   
 $x^\top \Sigma x = \frac{1}{4}(0.0225) + \frac{1}{4}(0.0225) + 2 \cdot \frac{1}{4}(0.0045) = 0.0135 \checkmark$



same mean, less spread: **diversification**. it comes from  $\rho < 1$  — with  $\rho = 1$ ,  $x^\top \Sigma x$  would stay 0.0225.

**The model** (Markowitz, 1952) — least risk, subject to an expected-return target  $r$ :

$$\begin{aligned} \min_x \quad & x^\top \Sigma x && \text{variance of the return (risk)} \\ \text{s.t.} \quad & \mu^\top x \geq r && \text{expected return meets the target} \\ & \mathbf{1}^\top x = 1, \quad x \geq 0 \quad x \in X \end{aligned}$$

the week 2 note's QP (its Section 5) with  $Q = 2\Sigma$ ,  $c = 0$ ; convex because  $\Sigma \succeq 0$  (its Section 6).

in the language of the bridge: the loss  $-\tilde{\xi}^\top x$  is random; Markowitz collapses it with two numbers — its variance (objective) and its mean (constraint).

**Data — the coding session**  $n = 3$ ,  $r = 0.10$

| asset $j$ | expected return $\mu_j$ | variance $\Sigma_{jj}$ | $\sigma_j = \sqrt{\Sigma_{jj}}$ |
|-----------|-------------------------|------------------------|---------------------------------|
| bond      | 0.06                    | 0.010                  | 0.10                            |
| tech      | 0.10                    | 0.040                  | 0.20                            |
| energy    | 0.15                    | 0.090                  | 0.30                            |

$$x = \begin{pmatrix} x_{\text{bond}} \\ x_{\text{tech}} \\ x_{\text{energy}} \end{pmatrix}, \quad \mu = \begin{pmatrix} 0.06 \\ 0.10 \\ 0.15 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0.010 & 0.001 & 0 \\ 0.001 & 0.040 & 0.015 \\ 0 & 0.015 & 0.090 \end{pmatrix}$$

covariances: bond–tech 0.001, tech–energy 0.015, bond–energy 0

written out term by term — exactly the lines typed in the code:

$$\begin{aligned} \min \quad & 0.010 x_{\text{bond}}^2 + 0.040 x_{\text{tech}}^2 + 0.090 x_{\text{energy}}^2 + 2(0.001) x_{\text{bond}} x_{\text{tech}} + 2(0.015) x_{\text{tech}} x_{\text{energy}} \\ \text{s.t.} \quad & 0.06 x_{\text{bond}} + 0.10 x_{\text{tech}} + 0.15 x_{\text{energy}} \geq 0.10 \\ & x_{\text{bond}} + x_{\text{tech}} + x_{\text{energy}} \leq 1 \\ & x_{\text{bond}}, x_{\text{tech}}, x_{\text{energy}} \geq 0 \end{aligned}$$

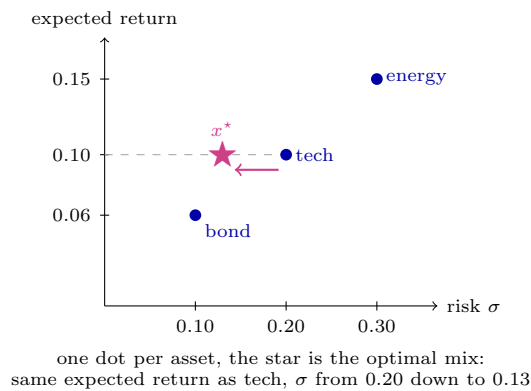
budget “ $\leq 1$ ”: leftover sits in cash earning 0 — never helps, so the sum is 1 at the optimum

**Solution**

$$\boxed{x^* \approx (0.39, 0.29, 0.31), \quad x^{*\top} \Sigma x^* \approx 0.0169, \quad \sigma \approx 0.130} \quad \mu^\top x^* = 0.10 \text{ — target met exactly}$$

all in tech: same expected return 0.10, but  $\sigma = 0.20$ . the mix cuts  $\sigma$  by about a third.

every asset gets a share — not a vertex (week 2 note, Section 5).



**demo** `week03_markowitz.ipynb` — Colab — build this model live from the skeleton, then compare with the completed cell.

**demo** `demos/markowitz_portfolio.py` — the same model as a plain script: prints the model, solves it, plots the weights  $x^*$ .

**takeaway** the running example: the loss  $-\tilde{\xi}^\top x$  stays; each later model changes only how it is collapsed to a number — today variance, next CVaR and VaR.

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**next** no class Sep 26. week 6 — duality: every LP comes with a shadow LP, and why  $\max = \min$ .